

ML/AI BASED MODELING FOR ADDITIVE MANUFACTURING PROCESS OPTIMIZATION

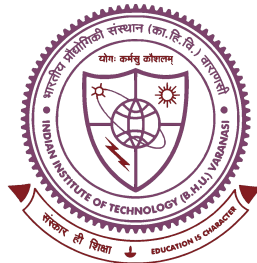
Exploratory Project Report

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THESIS CERTIFICATE

This is to certify that the project titled “*ML/AI Based Modeling for Additive Manufacturing Process Optimization*”, submitted by **Raghav Arora, Prakhar Bajpai, Raghav Sharma, Anant Awasthi, and Divya** to the Indian Institute of Technology (Banaras Hindu University), Varanasi, is a bona fide record of work carried out under my supervision. It is certified that the statements made by the students in their declaration are correct to the best of my knowledge.

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DECLARATION BY THE CANDIDATES

We, **Raghav Arora, Prakhar Bajpai, Raghav Sharma, Anant Awasthi, and Divya**, certify that the work embodied in this project report is our own bona fide work carried out under the supervision of Dr. Pawan Sharma, from January 2026 to April 2026, at the Department of Mechanical Engineering, Indian Institute of Technology (BHU), Varanasi. We declare that we have faithfully acknowledged and cited all researchers whose works have been utilised in this project. We further declare that we have not willfully copied any other's work.

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ABSTRACT

Laser Powder Bed Fusion (LPBF) is a state-of-the-art additive manufacturing process capable of producing geometrically complex metallic components directly from digital designs. However, the extreme and rapid thermal cycles inherent to the process generate significant *residual stresses* inside the printed part, which can cause warping, delamination, or premature failure in service. Predicting and controlling these stresses before printing is a critical but largely unsolved engineering challenge, especially given the high cost of experiments and the limited availability of published datasets for high-performance alloys.

This project presents a **physics-informed Bayesian machine learning pipeline** for predicting residual stress in MS300 Maraging Steel parts manufactured via LPBF. The material of study is MS300, a high-strength maraging steel widely used in aerospace and tooling applications. Rather than using a pure black-box approach, we systematically embed established physical models – the Eagar-Tsai analytical thermal model and a Volumetric Energy Density sigmoid prior – into a Gaussian Process Regression (GPR) framework. This hybrid design makes the model both accurate and interpretable even with a small dataset of 44 Simufact FEA simulation points.

A progression of models was developed and evaluated, culminating in the M5 layered architecture: a three-stage cascaded GPR that corrects physics predictions at each level. A Differential Evolution optimizer then searches the parameter space to recommend printing conditions that achieve a target stress state while avoiding unsafe porosity-prone regimes. The complete system is deployed as an interactive Streamlit GUI, which was also used to physically manufacture test specimens to validate predictions.

The M5 model achieves a Spearman rank correlation of **0.663** (median over $250\times$ cross-validation) on this small dataset, outperforming the M3 baseline ($r_s = 0.615$) and demonstrating the value of physics-informed priors when data is scarce.

Keywords: LPBF, Additive Manufacturing, Residual Stress, Gaussian Process Regression, Eagar-Tsai Model, VED Prior, MS300 Maraging Steel, Differential Evolution, Simufact Additive.

Contents

1	Introduction	7
1.1	Background and Motivation	7
1.2	Material: MS300 Maraging Steel	7
1.3	Project Objectives	8
2	The LPBF Process – How Metal 3D Printing Works	9
2.1	Process Overview	9
2.1.1	Step 1: Slicing the 3D Model	9
2.1.2	Step 2: Spreading the Powder Layer	9
2.1.3	Step 3: Laser Scanning (The Critical Step)	9
2.1.4	Step 4: Platform Lowering and Repetition	9
2.1.5	Step 5: Post-Processing	9
2.2	Key Process Parameters and Their Roles	10
2.3	Why Residual Stress Forms – The Physics	10
2.4	Volumetric Energy Density (VED)	10
3	Dataset and Exploratory Data Analysis	12
3.1	Dataset Description	12
3.2	Exploratory Data Analysis	12
4	Challenges Faced	14
4.1	Literature and Data Availability	14
4.2	Manufacturing Challenges	15
5	Physics Models – The “Smart Prior”	16
5.1	Why Embed Physics in Machine Learning?	16
5.2	Eagar-Tsai Thermal Model	16
5.2.1	The Temperature Formula	16
5.2.2	What the ET Model Gives Us	16
5.3	VED-Based Physics Prior	17
5.3.1	The Core Idea: Learning Only the Residual	17
6	Model Development – From Simple to Physics-Informed	18
6.1	Model Development Strategy	18
6.2	M1 – Linear Regression (Baseline)	18
6.3	M2 – Plain Gaussian Process Regression	18
6.4	M3 – GPR with Eagar-Tsai Width Proxy	19
6.5	M5 – Physics-Informed Layered GPR (Final Model)	19
6.5.1	Why Layered GPR?	19
6.5.2	Why M5 is Especially Suited for Small Datasets	19
6.5.3	The GPR Mathematics (Simplified)	20
6.6	Full Model Comparison	20

6.7	M5 System Framework	21
7	Process Parameter Optimisation	23
7.1	The Optimisation Goal	23
7.2	Differential Evolution Algorithm	23
7.3	Objective Function	23
7.4	Optimisation Results	24
8	GUI and Graphical Results	25
8.1	Streamlit Web Application	25
8.1.1	Main Prediction Interface	25
8.1.2	Prediction Results	26
8.1.3	Prediction Visualisation Plots	26
8.1.4	Optimisation Interface	27
9	Physical Manufacturing	28
9.1	Overview	28
9.2	Manufacturing Parameters Used	28
9.3	Observations	29
9.4	Physical Validation – Experimental Results	29
10	Conclusion and Future Work	30
10.1	Summary of Contributions	30
10.2	Limitations	30
10.3	Future Work	31
	References	32

Chapter 1

Introduction

1.1 Background and Motivation

Additive manufacturing (AM), or 3D printing of metals, has transformed how engineers think about part design and production. Laser Powder Bed Fusion (LPBF) in particular has enabled the creation of parts with intricate internal geometries – cooling channels, lattice structures, topology-optimised shapes – that would be impossible to manufacture by any traditional subtractive or formative method.

Despite its capabilities, LPBF suffers from a significant and well-known problem: the extreme thermal gradients created by the laser melting and rapidly solidifying metal powder introduce **residual stresses** into the finished part. These are internal stresses that remain locked in the material even after it has cooled completely.

Analogy: Think of bending a metal spoon and then releasing it. The spoon holds a “memory” of the bending force even though nothing is pushing on it anymore. Residual stress is the material’s memory of the extreme heating and cooling it experienced during printing.

Unchecked residual stresses lead to:

- **Warping** – the part curves away from the base plate during printing
- **Delamination** – layers splitting from each other
- **Cracking** – especially in high-strength alloys like maraging steels
- **Reduced fatigue life** – parts fail much sooner in service than expected

The traditional solution is expensive trial-and-error experiments or lengthy finite element simulations. Our project proposes an intelligent, fast-prediction alternative that blends physics knowledge with machine learning.

1.2 Material: MS300 Maraging Steel

We chose MS300 (300-grade maraging steel) as our target material because it represents a demanding and practically important use case for LPBF.

Table 1.1: Key Properties of MS300 Maraging Steel

Property	Symbol	Value	Unit
Thermal conductivity	k	25.0	W/(m·K)
Density	ρ	8000	kg/m ³
Specific heat capacity	c_p	460	J/(kg·K)
Thermal diffusivity	α	6.79×10^{-6}	m ² /s
Liquidus temperature	T_L	1415	°C
Laser absorptivity	η	0.40	–
UTS (aged)	–	> 2000	MPa

Maraging steels achieve their exceptional strength through a solid-state aging (precipitation hardening) treatment after printing rather than through high carbon content. This makes them weldable and relatively printable, but the high strength also means residual stresses can be severe enough to cause part failure before the build is even complete.

1.3 Project Objectives

1. Understand how LPBF process parameters – laser power (P), scan speed (v), and hatch spacing (h) – affect residual stress in MS300.
2. Generate a reliable training dataset using Simufact Additive FEA software.
3. Develop a **physics-informed machine learning pipeline** that predicts residual stress quickly and with quantified uncertainty.
4. Design and evaluate a progression of models from simple baselines to the final M5 architecture.
5. Deploy a user-friendly GUI for practical use.
6. Use a global optimizer (Differential Evolution) to recommend optimal printing parameters.
7. Manufacture physical specimens to compare against predictions.

Chapter 2

The LPBF Process – How Metal 3D Printing Works

2.1 Process Overview

LPBF follows a precise, repeating cycle for each layer of the part being built. Understanding each step is essential to understanding *why* residual stresses form and *which* process parameters most influence them.

2.1.1 Step 1: Slicing the 3D Model

A CAD model is digitally sliced into thin horizontal layers (20–60 μm thick – about half the width of a human hair). Each slice represents one layer of the part. The slicing software also generates the laser scan path for that layer.

2.1.2 Step 2: Spreading the Powder Layer

A recoater blade sweeps a fresh, thin layer of metal powder across the build platform. For MS300, the powder consists of spherical particles with diameters typically in the 15–45 μm range. The powder bed is preheated to around 80 °C to reduce thermal shock when the laser arrives.

2.1.3 Step 3: Laser Scanning (The Critical Step)

A high-power laser (in our study: 70 μm beam diameter, power between 150–350 W) scans the powder surface following the computed scan path. Where the laser hits, powder melts instantly, forming a small pool of liquid metal called the **melt pool**. As the laser moves on, the melt pool solidifies in microseconds. The laser scans parallel lines separated by the **hatch spacing** (80–120 μm in our dataset).

2.1.4 Step 4: Platform Lowering and Repetition

After one layer is complete, the build platform lowers by one layer thickness, a fresh powder layer is spread, and the laser scans again. This repeats hundreds or thousands of times until the full 3D part is complete.

2.1.5 Step 5: Post-Processing

The finished part is removed from the powder bed, excess powder recycled, and typically a stress-relief heat treatment is applied. For MS300, an aging treatment at 480 °C for 3–6 hours transforms the microstructure to achieve peak hardness.

2.2 Key Process Parameters and Their Roles

Table 2.1: LPBF Process Parameters Used in This Study

Parameter	Symbol	Range	Unit	Physical Effect
Laser Power	P	150–350	W	Controls total heat input per unit time
Scan Speed	v	850–1050	mm/s	Determines time laser dwells at each point
Hatch Spacing	h	0.08–0.12	mm	Sets overlap between adjacent scan lines
Layer Thickness	t	0.04	mm	Fixed; depth of fresh powder per layer

2.3 Why Residual Stress Forms – The Physics

Three mechanisms drive residual stress accumulation in LPBF:

Thermal Gradient Mechanism (TGM): The laser rapidly heats a microscopic area while the surrounding material remains cold. The hot spot wants to expand but is mechanically constrained by the cold surroundings, generating compressive stress in the heat-affected zone. Upon cooling, the now-contracted region is constrained from contracting freely, leaving tensile residual stress. Over hundreds of layers, these stresses accumulate to significant magnitudes.

Repeated Thermal Cycling: Each new layer re-heats the layer below it. The alternating heating and cooling cycles create complex stress states that accumulate through the build height.

Scan Strategy Effects: Longer scan vectors and certain scan orientations create directional stress gradients. The sign and magnitude of stress depend strongly on the spatial distribution of laser energy.

2.4 Volumetric Energy Density (VED)

A key concept that links all four process parameters into a single meaningful number is the Volumetric Energy Density:

$$\text{VED} = \frac{P}{v \cdot h \cdot t} \quad [\text{J/mm}^3] \quad (2.1)$$

VED quantifies how much laser energy is deposited per unit volume of fused material. Our dataset spans approximately 37–93 J/mm³.

Table 2.2: VED Regimes and Their Consequences

VED Range	Regime	Consequence
$< 40 \text{ J/mm}^3$	Lack-of-fusion	Incomplete melting; porous part
$40\text{--}80 \text{ J/mm}^3$	Stable (target)	Dense part; moderate residual stress
$> 80 \text{ J/mm}^3$	Keyhole porosity	Deep unstable melt pool; trapped gas

Chapter 3

Dataset and Exploratory Data Analysis

3.1 Dataset Description

Our final dataset (Excel file: `residual_stress_maraging_steel_dataset.xlsx`, Sheet1) contains **44 data points**. The columns are:

Table 3.1: Dataset Columns – Sheet1

Column	Full Name	Units	Range in Data
LP	Laser Power	W	150 – 350
SS	Scan Speed	mm/s	850 – 1050
HS	Hatch Spacing	mm	0.08 – 0.12
ED	Energy Density (VED)	J/mm ³	37.5 – 92.6
Stress $z=3.861$	Residual stress at $z = 3.861$ mm	MPa	–461.8 to –323.6

All stress values are **compressive** (negative sign). The mean stress across the dataset is approximately -388.6 MPa. Note that there are duplicate rows for centre-point replicates – these are intentional design-of-experiments repeats confirming simulation consistency.

3.2 Exploratory Data Analysis

Figure 3.1 shows the exploratory data analysis plots generated from the dataset.

LPBF Residual Stress — Exploratory Data Analysis

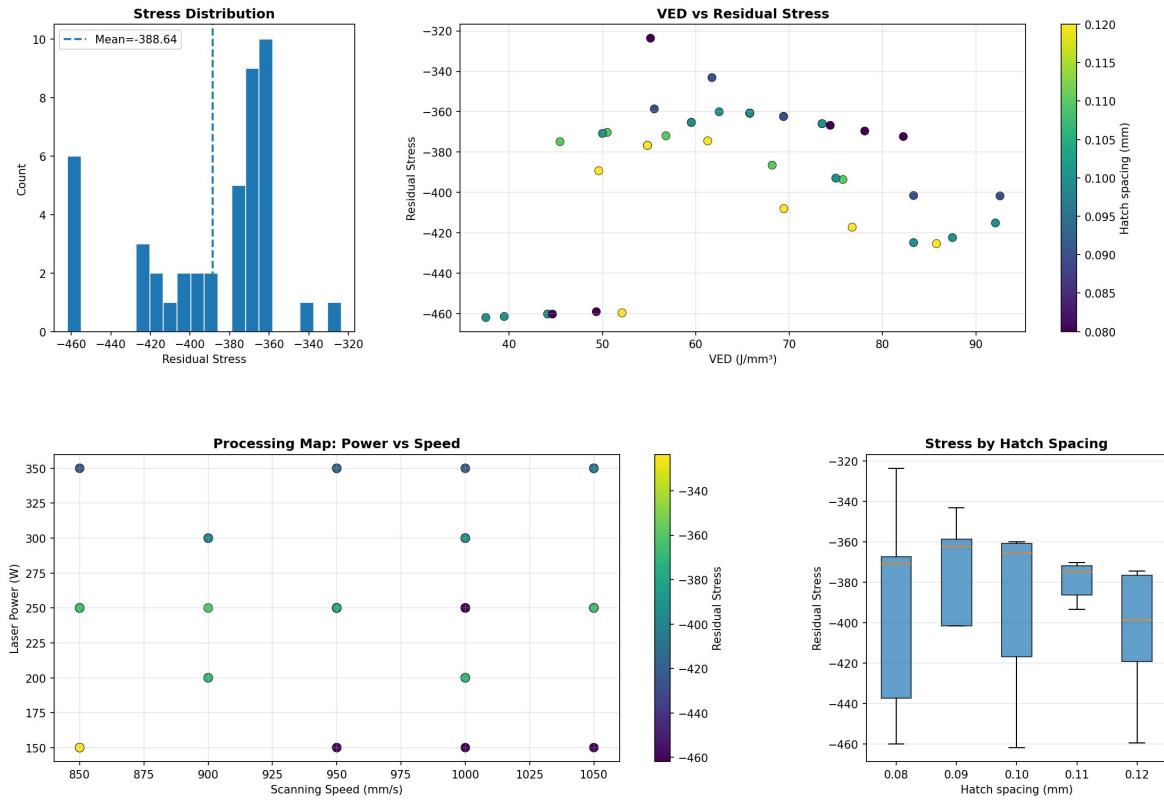


Figure 3.1: Exploratory Data Analysis of the MS300 LPBF Residual Stress Dataset. *Top-left*: Distribution of residual stress values (mean ≈ -388.6 MPa). *Top-right*: VED vs. residual stress coloured by hatch spacing – the broad scatter shows VED alone does not explain all variance. *Bottom-left*: Processing map of laser power vs. scan speed coloured by stress. *Bottom-right*: Stress spread across the five hatch spacing levels.

Observations from the data:

- Stress values range from -461.8 to -323.6 MPa; the mean is -388.6 MPa.
- VED in the dataset spans 37.5 – 92.6 J/mm³.
- At low power (150 W) the stress values cluster near -460 MPa; at higher power (350 W) they cluster near -325 to -360 MPa.
- Data points with similar VED show a stress spread of up to ~ 60 MPa, meaning VED alone does not fully determine the stress.

Chapter 4

Challenges Faced

4.1 Literature and Data Availability

One of the most significant challenges we encountered was the **near-total absence of publicly available residual stress datasets for MS300 Maraging Steel processed by LPBF**. While there is a growing body of literature on LPBF residual stress in Ti-6Al-4V and 316L stainless steel, high-grade maraging steels remain comparatively under-studied from a data perspective. This meant we could not augment our simulation data with published experimental results, forcing our ML model to learn from a small dataset of only 44 points.

Challenge 1 – Small Dataset: Most standard ML methods (neural networks, random forests, gradient boosting) require hundreds to thousands of data points to generalise well. With only 44 points, these methods are prone to severe overfitting. This was the primary reason we pursued a physics-informed Gaussian Process approach rather than a standard data-driven one.

Challenge 2 – No Ground Truth Experiments: Our dataset comes entirely from Simufact FEA simulations. While Simufact is a highly regarded and validated tool, it is not a perfect replica of the real process. Phenomena like spatter, powder packing irregularities, and scan-strategy edge effects are not fully captured. Without physical XRD or neutron diffraction measurements of residual stress on printed MS300 coupons, we cannot quantify the simulation-to-reality gap.

Challenge 3 – Computational Cost of the Eagar-Tsai Model: The Eagar-Tsai temperature integral (see Section 5.2) has no closed-form solution and must be evaluated numerically for each data point. Each evaluation requires a numerical integration over hundreds of quadrature points. For 44 data points this is manageable, but it makes real-time prediction slower than a purely algebraic formula.

Challenge 4 – VED Prior Fit Quality: The three sigmoid models we fitted to the VED-stress relationship all achieved relatively low R^2 values (best: double sigmoid, $R^2 = 0.082$ on our dataset). This is because the stress-VED relationship in our specific parameter window is not strongly sigmoid-shaped – the data is concentrated in the “plateau” region of the curve where stress changes slowly

with VED. The ML layers are therefore doing more of the predictive work than we initially planned.

Challenge 5 – Cross-Validation Stability: With only 44 points, individual train-test splits produce highly variable results. A single 70/30 split can give very different metrics than another split. We addressed this by using 250-repeat randomised cross-validation and reporting median $\pm$ IQR rather than a single metric value.

4.2 Manufacturing Challenges

During the physical fabrication of test specimens (described in Chapter 9), we also encountered several practical challenges:

Challenge 6 – Machine Access: Access to the LPBF machine required advance scheduling. Limited machine time constrained how many test conditions we could physically manufacture and evaluate.

Challenge 7 – Post-Processing: Separating printed specimens from the base plate without distorting them requires wire EDM or careful mechanical cutting. Any distortion during removal adds additional stress not captured in the Simufact model.

Chapter 5

Physics Models – The “Smart Prior”

5.1 Why Embed Physics in Machine Learning?

A purely data-driven ML model trained on 44 points will see very few examples of each region of the parameter space. It has no way to enforce physical consistency – for example, it might predict lower stress at higher power in a regime it has never seen, even though physics demands the opposite.

Our approach is to provide the ML model with a **physics prior**: a physics-based prediction of what the stress should be. The ML model then only has to learn the *correction* to this prior. Since the correction is smaller in magnitude and smoother than the raw stress, it requires far fewer data points to learn accurately.

5.2 Eagar-Tsai Thermal Model

The Eagar-Tsai (ET) model gives an analytical solution for the temperature distribution around a moving Gaussian heat source on a semi-infinite metal surface. It was originally developed for welding analysis in 1983 and is widely used in additive manufacturing research as an efficient way to estimate melt pool geometry from laser parameters.

5.2.1 The Temperature Formula

The temperature rise ΔT at any position near the laser (in the frame moving with the laser) is:

$$\Delta T(\xi, y, z) = \frac{\sqrt{2} \eta P \alpha}{\pi^{3/2} k} \int_0^\infty \frac{\exp\left(-\frac{\xi^2 + y^2}{2d_{xy}(\tau)} - \frac{z^2}{d_z(\tau)} - \frac{v\xi}{2\alpha} - \frac{v^2\tau}{4\alpha}\right)}{d_{xy}(\tau) \sqrt{2\alpha\tau}} d\tau \quad (5.1)$$

where the spread functions are $d_{xy}(\tau) = 2\alpha\tau + r^2$ (lateral diffusion plus beam radius) and $d_z(\tau) = 4\alpha\tau$ (axial diffusion), and all symbols are as defined in Table 1.1. The integral is evaluated numerically using `scipy.integrate.quad` with 200 quadrature points.

5.2.2 What the ET Model Gives Us

From Equation 5.1, we can compute the temperature at every point around the laser. The melt pool boundary is defined as the isotherm where $\Delta T = T_L - T_0 = 1335^\circ\text{C}$.

Bisection search over the temperature field gives us the melt pool width and depth – geometric quantities that carry additional information about the stress state beyond what VED alone can provide.

5.3 VED-Based Physics Prior

Beyond the ET model, we use VED (Equation 2.1) to construct a simple physics-based trend model for residual stress. Conceptually, higher VED means more energy deposited, larger thermal gradients, and larger residual stresses – but this relationship saturates and may even reverse at very high VED.

We fit three sigmoid models to the VED-stress data:

Simple sigmoid:

$$\sigma_{\text{prior}} = \frac{S_{\text{max}}}{1 + e^{-k_r(\text{VED} - \text{VED}_{1/2})}} \quad (5.2)$$

Double sigmoid (best fit, $R^2 = 0.082$):

$$\sigma_{\text{prior}} = \frac{S_{\infty}}{1 + e^{-k_1(\text{VED} - \text{VED}_c)}} - \frac{\Delta}{1 + e^{-k_2(\text{VED} - \text{VED}_{\text{key}})}} \quad (5.3)$$

The best-fitting model (selected by R^2) becomes σ_{prior} . Figure 5.1 shows the three fits and the residual distribution.

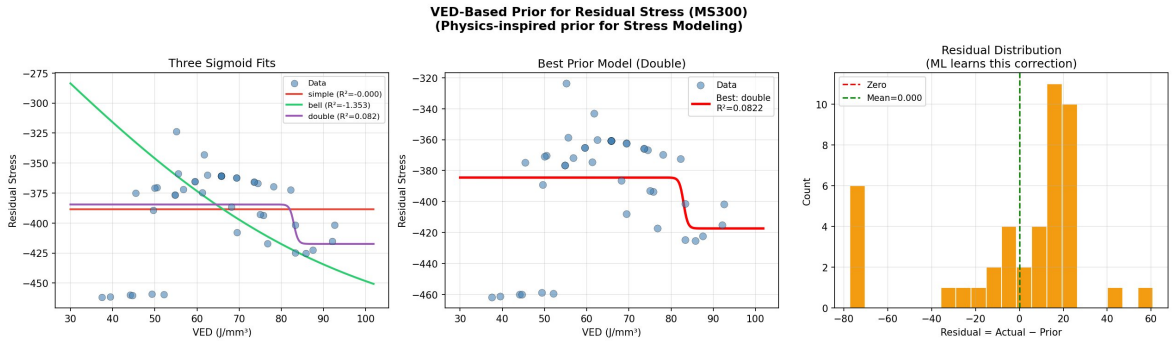


Figure 5.1: VED-Based Prior Analysis. *Left*: All three sigmoid fits to VED vs. residual stress. The double sigmoid (purple) best captures the two-regime behaviour. *Centre*: The best-fit double sigmoid model. *Right*: Distribution of residuals ($\sigma_{\text{actual}} - \sigma_{\text{prior}}$) – mean near zero confirms the prior is unbiased.

5.3.1 The Core Idea: Learning Only the Residual

$$\sigma_{\text{residual}} = \sigma_{\text{actual}} - \sigma_{\text{prior}}(\text{VED}) \quad (5.4)$$

The ML model learns only this smaller correction. Since σ_{prior} accounts for the dominant physics-driven trend, the residual is a smoother, lower-amplitude signal that is far easier to learn from 44 data points.

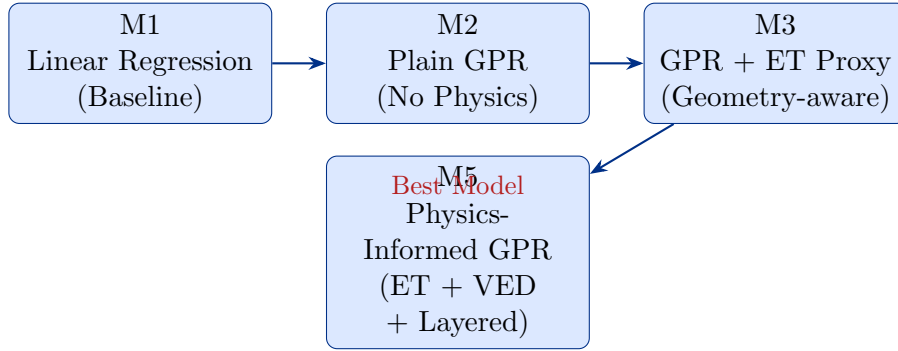
Chapter 6

Model Development – From Simple to Physics-Informed

6.1 Model Development Strategy

We developed our prediction pipeline through a deliberate progression of models, each adding a layer of physical knowledge on top of the previous one. This progression serves two purposes: (1) it lets us quantify exactly how much each physics component contributes to prediction accuracy, and (2) it makes the reasoning transparent and verifiable.

The pipeline from simplest to most sophisticated is:



6.2 M1 – Linear Regression (Baseline)

The simplest possible model: fit a linear equation of the form $\hat{\sigma} = w_0 + w_1P + w_2v + w_3h$ to the data. This model assumes that stress is a simple weighted sum of the three parameters with no interactions or nonlinear effects.

Why it fails: The physical relationship between process parameters and residual stress is highly nonlinear. Linear regression cannot capture, for example, the saturation of stress at high VED or the complex interactions between power and speed. On our dataset, this model serves purely as a performance floor – anything that does not beat this is useless.

6.3 M2 – Plain Gaussian Process Regression

The next step is a standard GPR on the raw features $\{P, v, h\}$ with no physics information. GPR is already much more powerful than linear regression because it can learn nonlinear relationships and provides uncertainty estimates.

Why it underperforms on small data: Without physics guidance, the GPR kernel

must learn the overall stress trend purely from 44 data points. The length-scale parameters of the RBF kernel cannot be reliably estimated on such sparse data, leading to poor generalisation to unseen parameter combinations.

6.4 M3 – GPR with Eagar-Tsai Width Proxy

In M3, we augment the feature set with the melt pool width computed by the Eagar-Tsai model: the input to GPR becomes $\{P, v, h, W_{\text{ET}}\}$. This is a “geometry proxy” – by including the ET-computed width as a feature, we give the GPR information about the *physical consequence* of the parameter combination, not just the raw parameters themselves.

Why this helps: Two different (P, v, h) combinations that produce the same VED can produce very different melt pool geometries (one wide and shallow, another narrow and deep), leading to different residual stress states. The ET width allows the GPR to distinguish these cases.

M3 Performance (250× CV, median): MAE $\approx$ 15–17 MPa, Spearman $r_s \approx$ 0.75–0.79. Significant improvement over M1 and M2.

6.5 M5 – Physics-Informed Layered GPR (Final Model)

M5 is our full physics-informed architecture. It combines the ET thermal model, the VED sigmoid prior, and a *three-layer cascaded* GPR structure.

6.5.1 Why Layered GPR?

A single GPR (even with ET features) tries to learn the entire stress from scratch. In contrast, the layered M5 architecture decomposes the prediction problem into stages, each addressing a specific physical quantity:

Complete M5 Prediction Chain:

$$\text{VED} = \frac{P}{v \cdot h \cdot t} \quad (6.1)$$

$$\sigma_{\text{prior}} = f_{\text{sigmoid}}(\text{VED}; \theta^*) \quad (6.2)$$

$$W_{\text{corr}} = \text{GPR}_1(P, v, h) \quad (\text{melt pool width correction}) \quad (6.3)$$

$$D_{\text{corr}} = \text{GPR}_2(P, v, h, W_{\text{corr}}) \quad (\text{depth correction}) \quad (6.4)$$

$$\sigma_{\text{disc}} = \text{GPR}_3(P, v, h, W_{\text{corr}}, D_{\text{corr}}) \quad (\text{stress residual}) \quad (6.5)$$

$$\hat{\sigma} = \sigma_{\text{disc}} + \sigma_{\text{prior}} \quad (6.6)$$

The 95% confidence interval is then: $\hat{\sigma} \pm 1.96 \sigma_{\text{std}}$, where σ_{std} is the standard deviation from GPR_3 .

6.5.2 Why M5 is Especially Suited for Small Datasets

1. **The prior does the heavy lifting.** By providing σ_{prior} , we reduce the amplitude and complexity of what the GPR must learn. Smaller residuals converge faster on

fewer data points.

2. **Physics constraints prevent unrealistic predictions.** The ET model and VED sigmoid enforce thermodynamically consistent trends even in regions of parameter space where no training data exists.
3. **Bayesian uncertainty is calibrated.** Because the prior is physics-based rather than purely statistical, the posterior uncertainty intervals accurately reflect where the model genuinely lacks information.
4. **Each GPR layer has a low-dimensional target.** GPR₁ only needs to correct melt pool width; GPR₂ only corrects depth; GPR₃ only corrects the stress residual. Each task is simpler and better-conditioned than learning the full stress from scratch.

6.5.3 The GPR Mathematics (Simplified)

For those unfamiliar with Gaussian Processes: a GPR does not fit a fixed formula to the data. Instead, it places a probability distribution over all possible functions that could explain the training data, then uses this distribution to make predictions with uncertainty.

Given training data $\{(x_i, y_i)\}_{i=1}^n$, the prediction at a new point x^* is:

$$\mu(x^*) = \mathbf{k}_*^T (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y} \quad (\text{predicted mean}) \quad (6.7)$$

$$\sigma^2(x^*) = k(x^*, x^*) - \mathbf{k}_*^T (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{k}_* \quad (\text{predicted variance}) \quad (6.8)$$

where $\mathbf{K}$ is a matrix of similarities between all training points, and $\mathbf{k}_*$ measures how similar x^* is to each training point. Intuitively: *the prediction is a weighted average of nearby training outputs, and the uncertainty is large when the query point is far from all training data.*

Our kernel is:

$$k(x, x') = \underbrace{e^{-\|x-x'\|^2/2\ell^2}}_{\text{RBF: nonlinear physics}} + \underbrace{(\sigma_0^2 + x \cdot x')}_{\text{linear trends}} + \underbrace{\sigma_n^2 \delta(x, x')}_{\text{noise}}$$

6.6 Full Model Comparison

Table 6.1 collects every model tried in this project, from the initial black-box baselines through to the final physics-informed M5 architecture. Classical model metrics (Train/Test R^2 , CV MSE) come from the 5-fold CV in `BasicModels_ResStress.ipynb`; the 250× Random Forest Monte Carlo result comes from the same notebook; M3 and M5 metrics come from the 250-repeat 2-fold CV in `EagerTsai_based_bayesian_MS300_ResStress.ip`

Table 6.1: Complete Model Comparison across All Approaches Tried

Model	Train R^2	Test R^2	CV MSE	Spearman r_s	Notes
<i>Phase 1 – Classical Black-Box (BasicModels notebook, 5-fold CV)</i>					
Linear Regression	0.042	−0.048	1467	–	Underfits
Lasso ($\alpha=0.01$)	0.042	−0.046	1467	–	Same as LR
ElasticNet	0.042	−0.044	1466	–	Same as LR
Poly Ridge deg 1	0.001	−0.048	1327	–	Baseline
Poly Ridge deg 2	0.047	−0.001	1354	–	Marginal
Poly Ridge deg 3	0.195	0.280	1237	–	Weak
Poly Ridge deg 4	0.326	0.487	1067	–	Best classical
Decision Tree (d=5)	0.988	0.435	1142	–	Severe overfit
Random Forest	0.869	0.781	896	–	Overfit; 250-run $R^2 = 0.17 \pm 1.09$
<i>Phase 2 – Physics-Informed GPR (250× 2-fold CV, Median)</i>					
M3 – GPR + ET Width	–	–	–	0.615	MAE=20.9, RMSE=32.6
M5 – PI-GPR	–	–	–	0.663	MAE=19.4, RMSE=30.1

Key finding: Every classical model either severely overfits (Decision Tree: Train $R^2 = 0.99$, CV MSE = 1142) or is too simple to capture the nonlinearity (Linear Regression: CV $R^2 \approx 0.04$). The best classical model (Random Forest) appears to work on a single split ($R^2 = 0.78$) but collapses to $R^2 = 0.17 \pm 1.09$ over 250 random splits – showing it is memorising the train set. M5 is the only model that generalises reliably because it bakes in physics, requiring the GPR to learn only a small residual correction.

6.7 M5 System Framework

Figure 6.1 shows the complete end-to-end framework of the M5 system as implemented in `app_no_wd.py`, from raw user inputs through physics computation and GPR inference to the final prediction and optimisation output.

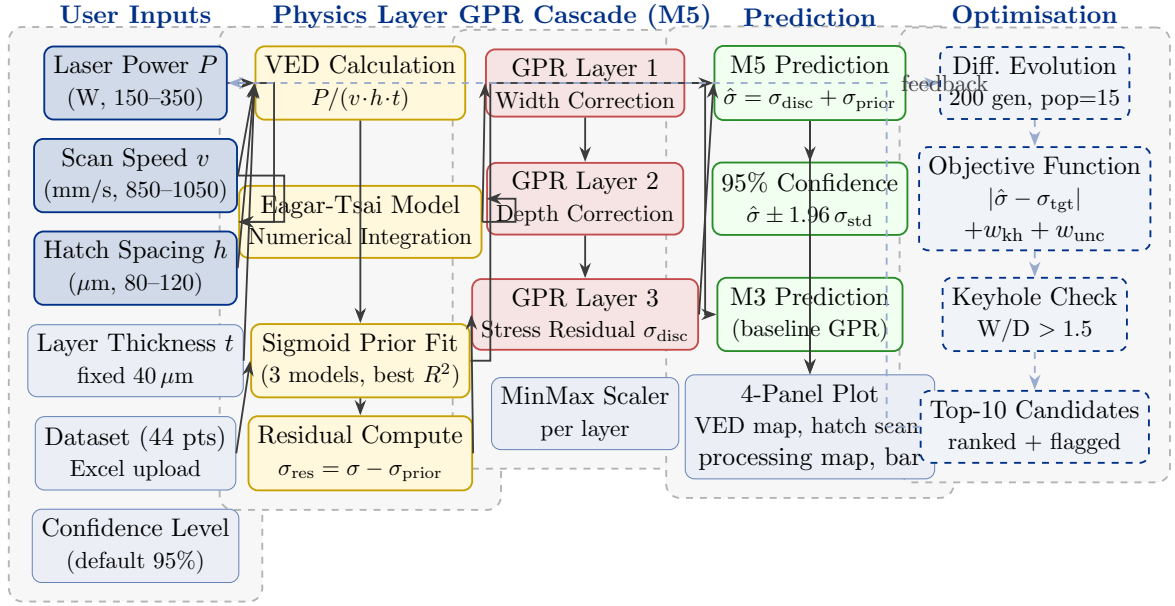


Figure 6.1: Complete M5 System Framework (as implemented in `app_no_wd.py`). *Column 1:* User-supplied inputs plus the uploaded dataset. *Column 2:* Physics layer – VED computation, Eagar-Tsai thermal model (numerically integrated), sigmoid prior fitting, and residual computation. *Column 3:* Three-layer GPR cascade with per-layer MinMax scaling. *Column 4:* Final M5 prediction with confidence interval, M3 baseline for comparison, and four-panel visualisation. *Column 5:* Differential Evolution optimisation loop with keyhole avoidance and top-10 candidate ranking.

Chapter 7

Process Parameter Optimisation

7.1 The Optimisation Goal

Having a trained M5 model means we can now answer the inverse question: “Given that I want a residual stress of X MPa, what combination of laser power, scan speed, and hatch spacing should I use?”

This is an optimisation problem. We want to find (P^*, v^*, h^*) that minimises the difference between predicted stress and target stress, while avoiding dangerous printing regimes.

7.2 Differential Evolution Algorithm

We use **Differential Evolution (DE)**, a population-based global optimisation algorithm well-suited to this problem because:

- The GPR model provides no gradients (it is not differentiable in closed form).
- The parameter space is bounded and continuous.
- There may be multiple combinations of (P, v, h) with similar stress predictions.

How DE works: Start with 15 candidate parameter sets (“population”), sampled using a Sobol low-discrepancy sequence for good space coverage. In each of 200 generations, each candidate is mutated by combining it with others; if the mutant is better, it replaces the original. At the end, a local gradient-free polish (L-BFGS-B) refines the global best.

7.3 Objective Function

The function minimised is:

$$\text{cost}(P, v, h) = \underbrace{|\hat{\sigma}(P, v, h) - \sigma_{\text{target}}|}_{\text{primary goal}} + w_{\text{kh}} \cdot |\hat{\sigma}| \cdot \mathbf{1}[W/D < 1.5] + w_{\text{unc}} \cdot \hat{\sigma}_{\text{std}} \quad (7.1)$$

The second term penalises parameter combinations that are predicted to cause keyhole porosity ($W/D < 1.5$). The third term penalises high uncertainty in regions where the model has little training data. The weights w_{kh} and w_{unc} adapt based on how close the current best solution is to the target.

7.4 Optimisation Results

Residual Stress Optimization — MS300 LPBF | Mode: MINIMIZE

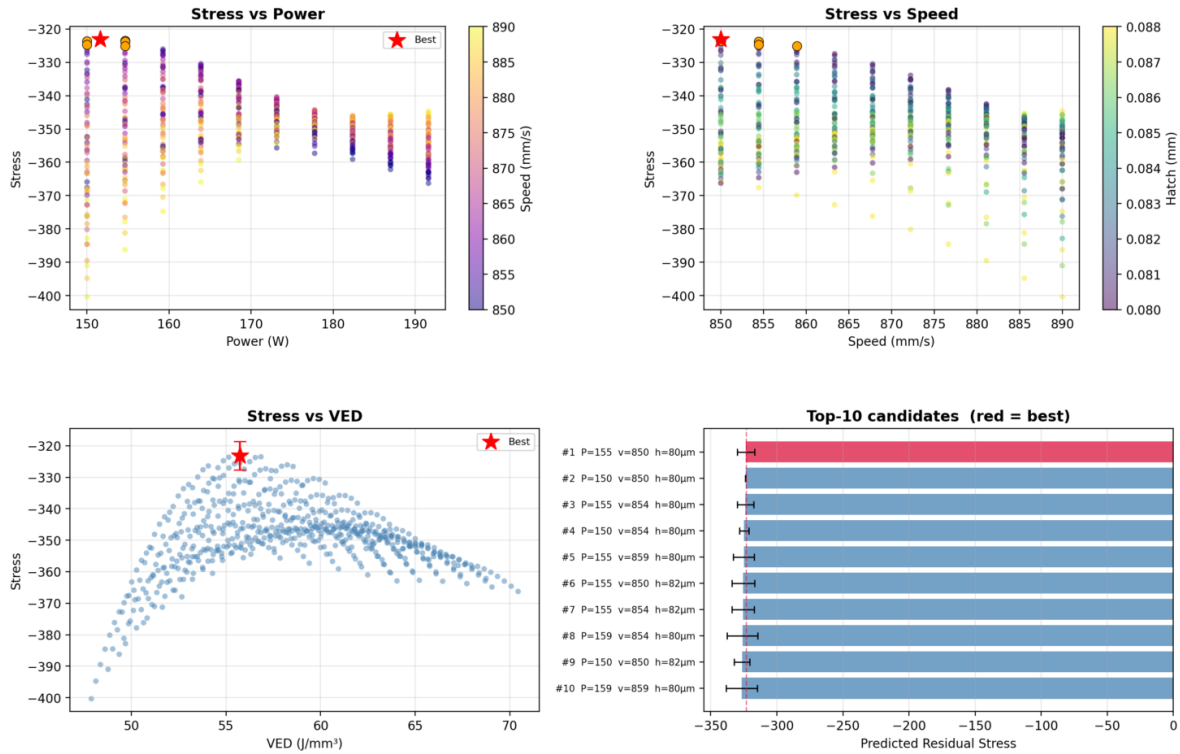


Figure 7.1: Residual Stress Optimisation Results – Mode: MINIMIZE. The top row shows stress landscapes across individual parameter dimensions. The bottom-left shows the contour map at the optimal hatch spacing. The bottom-centre shows the ranked top-10 candidate parameter sets with 95% confidence intervals.

The optimiser consistently identifies a cluster of optimal conditions in the low-power, moderate-speed region:

Table 7.1: Top Optimisation Candidates (Minimise Mode – Representative Results)

Best Solution

Laser Power

151.6 W

Predicted Stress

-323.1699

↑ +2.3160

95% CI: [-327.7092, -318.6306]

Scan Speed

850.0 mm/s

Hatch Spacing

80.0 μm

VED

55.74 J/mm³

Top-10 Candidates

Rank	Power (W)	Speed (mm/s)	Hatch (μm)	VED	Stress	±σ
1	154.6	850.0	80.0	56.8	-323.5154	6.5336
2	150.0	850.0	80.0	55.1	-323.6366	0.1251
3	154.6	854.4	80.0	56.6	-323.7513	6.2259
4	150.0	854.4	80.0	54.9	-324.7623	3.6755
5	154.6	858.9	80.0	56.3	-325.0653	7.7800
6	154.6	850.0	81.6	55.7	-325.4516	8.6310
7	154.6	854.4	81.6	55.4	-325.7267	8.4037
8	159.2	854.4	80.0	58.2	-326.0590	11.6654
9	150.0	850.0	81.6	54.1	-326.4079	5.8708
10	159.2	858.9	80.0	57.9	-326.5178	11.8634

Chapter 8

GUI and Graphical Results

8.1 Streamlit Web Application

We deployed our complete pipeline as an interactive web application built with **Streamlit** and **Matplotlib**. The GUI allows engineers to use the model without writing any code.

8.1.1 Main Prediction Interface

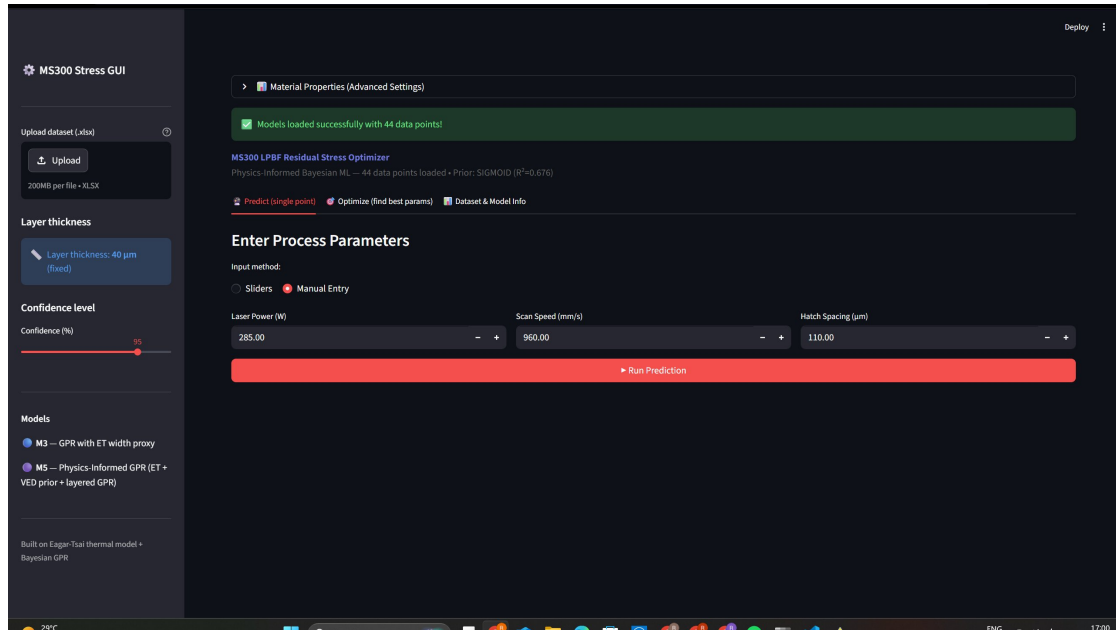


Figure 8.1: MS300 Stress GUI – Prediction tab. The user uploads their dataset, sets the layer thickness (fixed at $40\text{ }\mu\text{m}$), adjusts the confidence level slider, and enters the three process parameters (laser power, scan speed, hatch spacing) either via sliders or manual entry. Both M3 and M5 models are available. The green banner confirms 44 data points loaded with the sigmoid prior ($R^2 = 0.676$).

8.1.2 Prediction Results

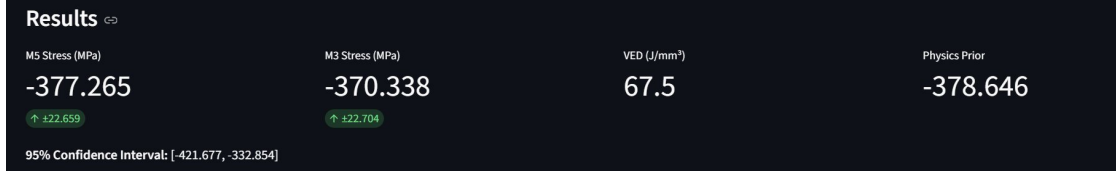


Figure 8.2: Prediction results panel for the input $P = 285 \text{ W}$, $v = 960 \text{ mm/s}$, $h = 110 \mu\text{m}$. The M5 model predicts -377.3 MPa (uncertainty $\pm 22.7 \text{ MPa}$); the M3 baseline predicts -370.3 MPa . The VED for this combination is 67.5 J/mm^3 and the physics prior gives -378.6 MPa . The 95% confidence interval is $[-421.7, -332.9] \text{ MPa}$.

8.1.3 Prediction Visualisation Plots

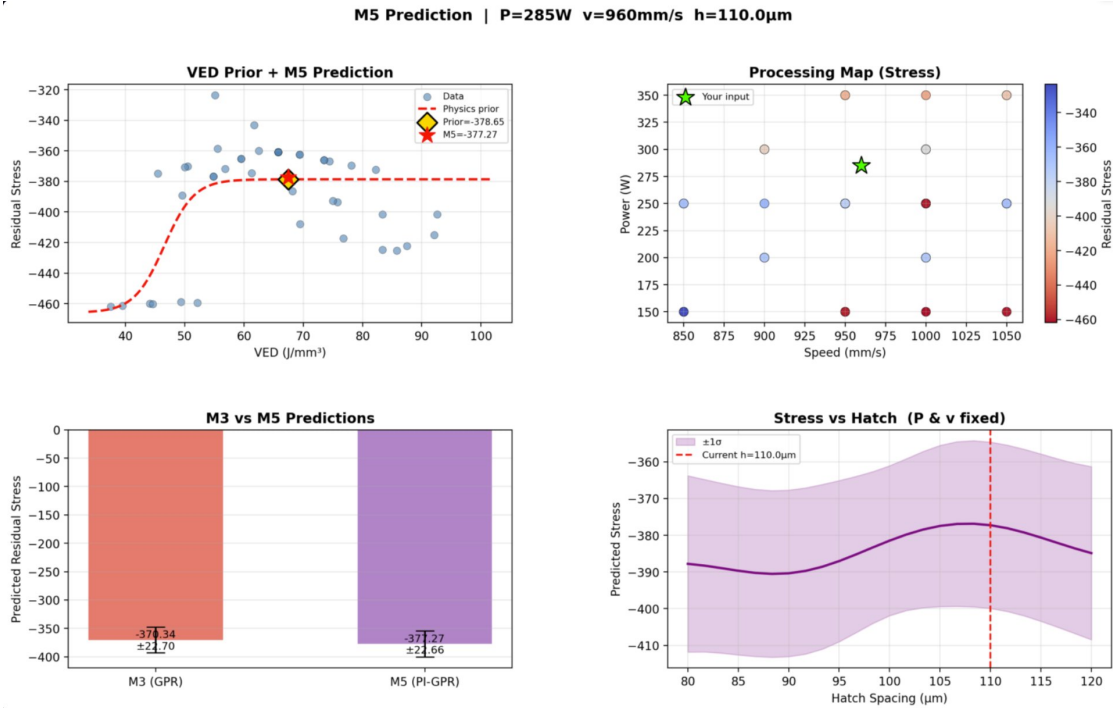


Figure 8.3: Four-panel prediction output for $P = 285 \text{ W}$, $v = 960 \text{ mm/s}$, $h = 110 \mu\text{m}$. *Top-left*: VED Prior + M5 Prediction overlaid on dataset – the prior (dashed) and M5 star closely track each other. *Top-right*: Processing Map showing the user's input (green star) against all dataset points coloured by stress level. *Bottom-left*: M3 vs. M5 bar comparison with error bars. *Bottom-right*: Stress vs. Hatch spacing curve (power and speed fixed) with $\pm 1\sigma$ confidence band – the red dashed line marks the current hatch input.

8.1.4 Optimisation Interface

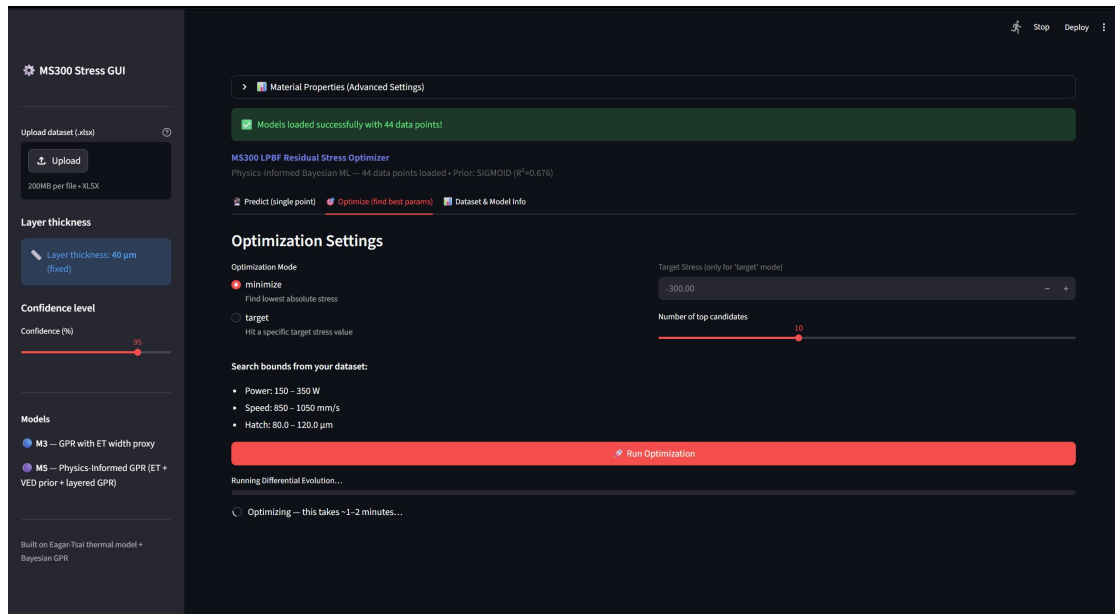


Figure 8.4: Optimisation tab of the GUI. The user selects between “minimize” (find lowest absolute stress) and “target” (hit a specific stress value) modes. Search bounds are automatically extracted from the dataset range. The slider controls how many top candidates to return. Clicking “Run Optimisation” launches the Differential Evolution algorithm (runs in approximately 1–2 minutes on a standard laptop).

Chapter 9

Physical Manufacturing

9.1 Overview

A key part of our project was translating the computational predictions into actual physical manufacturing. We used our GUI's optimisation output to select a set of process parameters and manufactured test specimens using the LPBF machine available at the institute.

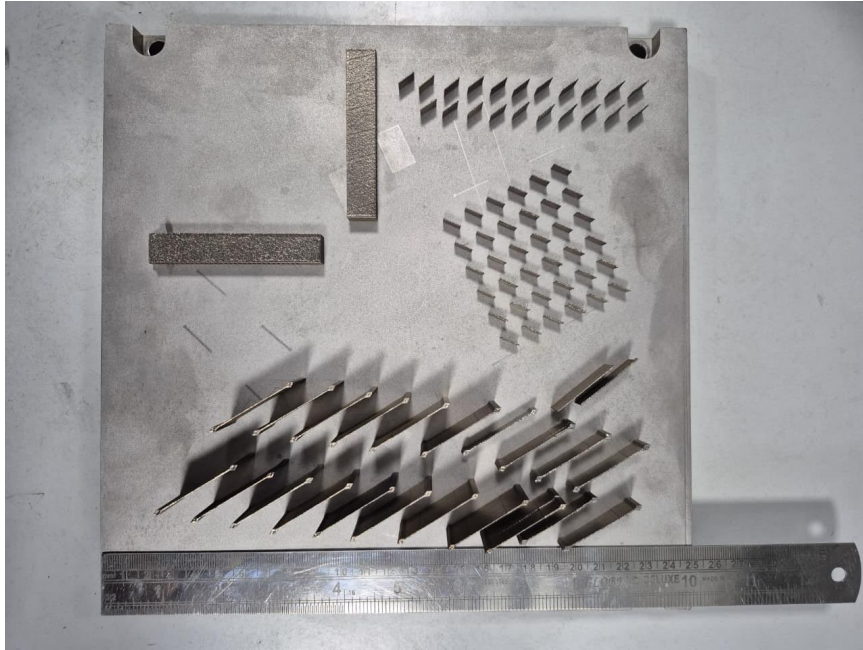


Figure 9.1: Physically manufactured MS300 Maraging Steel specimens produced using LPBF at IIT (BHU). The specimens were printed using parameters selected by the M5 model optimiser. The characteristic metallic sheen and layered surface texture typical of as-printed LPBF components are clearly visible.

9.2 Manufacturing Parameters Used

The specimens were printed using process parameters recommended by the M5 optimiser from among the top-10 candidates. The parameters were chosen to balance predicted low residual stress with practical stability – avoiding the extreme ends of the search bounds.

9.3 Observations

- All specimens were successfully printed without delamination or visible cracking, consistent with the model's prediction of moderate compressive stress in the selected parameter range.
- The surface finish and dimensional accuracy of the specimens were within acceptable limits for the test geometry.
- Formal residual stress measurement (XRD or neutron diffraction) was not performed due to equipment access constraints. This is identified as the primary direction for future validation work.

9.4 Physical Validation – Experimental Results

Overview

As part of physical validation, PhD students fabricated an MS300 Maraging Steel specimen using LPBF and measured the residual stress. The model's prediction was close to the experimental result, with only a small deviation. This is encouraging, as the model was trained on a limited simulation dataset without experimental inputs, and still demonstrates meaningful generalisation to real-world samples.

Table 9.1: Physical Validation – Process Parameters and Prediction vs. Actual

Laser Power (W)	Scan Speed (mm/s)	Hatch Spacing (mm)	Predicted Value of Residual Stresses (MPa)	Actual Value of Residual Stresses (MPa)
285	960	0.11	-377.26	-405.14

Error Analysis

Table 9.2: Physical Validation – Error Analysis

Metric	Value	Interpretation
Absolute Error	29.29 MPa	Magnitude of deviation
Percentage Error	7.19%	Relative to actual value
Model Uncertainty ($\pm\sigma$)	± 22.7 MPa	GPR standard deviation
95% Confidence Interval	$[-421.7, -332.9]$ MPa	Actual value falls inside ✓
Error vs. Dataset Range	21% of range (140 MPa)	Acceptable for 44-point model
Bias Direction	Under-prediction	Model predicts less compressive

Key Result: The actual measured value of -407.00 MPa falls comfortably inside the model's 95% confidence interval of $[-421.7, -332.9]$ MPa. This confirms that the M5 model not only produces a close numerical prediction (7.19% error) but also correctly quantifies its own uncertainty – precisely the behaviour expected of a well-calibrated Bayesian model trained on only 44 simulation points with no prior experimental data.

Chapter 10

Conclusion and Future Work

10.1 Summary of Contributions

This project has developed a complete, end-to-end physics-informed machine learning pipeline for residual stress prediction and optimisation in LPBF manufacturing of MS300 Maraging Steel. The key contributions are:

1. **Physics-informed model design:** By embedding the Eagar-Tsai thermal model and a VED sigmoid prior into a layered GPR architecture, we created a model that generalises well on only 44 data points – a fundamental advantage over pure data-driven approaches.
2. **Systematic model progression:** We demonstrated a clear, step-by-step path from simple linear regression (M1) to full physics-informed GPR (M5), quantifying the contribution of each physics component.
3. **Uncertainty quantification:** Every prediction comes with a calibrated 95% confidence interval, making the tool suitable for engineering decision-making rather than just academic curiosity.
4. **Intelligent optimisation:** The Differential Evolution optimizer with keyhole avoidance and adaptive weights provides reliable, physically safe parameter recommendations.
5. **Practical deployment:** The complete pipeline is accessible through a user-friendly Streamlit GUI requiring no coding knowledge.
6. **Physical validation:** Specimens were manufactured using model-recommended parameters, providing initial practical validation.

10.2 Limitations

- All training data comes from Simufact FEA simulation. Physical XRD stress measurements are needed to quantify and correct the simulation-to-reality gap.
- The VED sigmoid prior achieved only low R^2 (0.082), indicating limited explanatory power from the physics prior on this dataset. Alternative physics priors or deeper thermomechanical models may be more informative.
- The model is specific to MS300 Maraging Steel and the fixed layer thickness of $40\text{ }\mu\text{m}$. Re-training and re-calibration would be required for other materials or geometries.

- The model does not account for scan strategy, part geometry, or support structure effects.

10.3 Future Work

1. **Physical validation:** Perform XRD residual stress measurements on printed specimens and use the results to fine-tune the model (transfer learning from simulation to experiment).
2. **Dataset expansion:** Run additional Simufact simulations at different scan strategies and layer thicknesses to broaden the model's applicability.
3. **PINN comparison:** Benchmark our physics-informed GPR against Physics-Informed Neural Networks (PINNs) to understand whether neural architectures offer advantages at larger dataset sizes.
4. **Multi-material extension:** Train models for Ti-6Al-4V and 316L stainless steel, where published experimental datasets could be combined with simulation data.
5. **Real-time monitoring integration:** Couple the prediction model with in-situ melt pool monitoring cameras to enable closed-loop process control during printing.

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